

Proposal

Project Title

BiteFlow:

An Integrated Cloud-Based Ecosystem for Food Ordering and Logistics Management

Problem Description

The contemporary food service industry faces significant operational challenges due to fragmented communication between customers, restaurants, and delivery personnel.

Traditional ordering methods often suffer from:

Operational Inefficiency: Manual order processing is prone to human error, leading to incorrect orders and financial losses.

Lack of Real-Time Visibility: Customers and restaurant managers lack transparency regarding the order lifecycle, from preparation to "last-mile" delivery.

Scalability Constraints: Small to medium-sized eateries struggle to manage high-volume traffic and delivery logistics without a centralized automated system.

Data Fragmentation: Without a unified platform, restaurants lose valuable insights into customer behavior and inventory trends, hindering strategic growth.

System objectives and description

The system aims to:

- Automate the entire order lifecycle.: This includes order placement, payment processing, and delivery assignment to cut down on human errors.
- Provide real-time tracking of orders and delivery status. This increases transparency and builds user trust.
- Improve communication between customers, restaurants, and delivery partners with instant notifications and updates.
- Help restaurants manage incoming orders effectively. This reduces preparation and delivery delays.
- Enhance the user experience with a simple, fast, and user-friendly interface.

- Support growth to manage many users and orders without losing performance.
- - Provide basic analytics like order history, delivery time tracking, and customer activity to help with better decision-making.
- - The system also ensures food is delivered perfectly, without mixing or changing shape due to road conditions.
- - Clean energy and sustainability are utilized in food delivery via electric motorcycles.
- - An algorithm is integrated to ensure the best route based on traffic congestion.
- - The customer is notified that the order has been delivered via confirmation button, and service ratings are requested.
- - The delivery person's arrival is also confirmed via their address, and confirmation is requested from them as well.

Stakeholders

The BiteFlow system includes several stakeholders who interact with the system at different levels.

Primary Stakeholders:

Customers Browse restaurants, place orders, make payments, and track deliveries in real time.

Restaurant Staff / Managers: Manage menus, receive and process orders, and oversee food preparation.

Delivery Partners: Accept delivery requests, pick up orders, and deliver them to customers

Secondary Stakeholders:

System Administrators: Manage users, monitor system performance, handle accounts, and ensure system security.

Payment Service Providers: External services that handle secure online payment transactions.

Map Service Providers: External systems used for location tracking, navigation, and route optimization.

cloud providers: like Amazon

Relations to System Models:

The defined system objectives and stakeholders will be directly used in developing the system models.

- Stakeholders will be shown as actors in the Use Case Diagram. This diagram will illustrate how each user interacts with the system.

- The system objectives will help identify Functional Requirements. These requirements will later be represented in the Use Case and Activity Diagrams.

- Real-time ordering, tracking, order processing and delivering confirmation for both sides objectives will be displayed in the Activity Diagram to show the system workflow.

- External stakeholders, like payment and map services, will be included in the Context Diagram. Here, they will be shown as external systems interacting with BiteFlow. This approach ensures consistency between the system requirements and the system design models.